



HealthPulse Presentation Script (Final Version)

(You can read this verbatim in your presentation)



PM – Project Introduction (1 minute)

“Hello everyone, and thank you for joining us.

Today, we’re presenting the HealthPulse No-Show Prediction and Patient Flow Optimization initiative. As the Project Manager, I led the coordination of our cross-functional team — Data Engineering, Data Science, and Data Analytics — working alongside our Business Analyst to deliver a solution that improves clinic performance, reduces no-shows, and enhances patient experience.

We managed this project in a one-week focused sprint using Trello for visibility, daily stand-ups for alignment, and structured checkpoints to ensure quality. Our goal was to transition HealthPulse from reactive operations to proactive, data-driven scheduling.

I’ll now hand over to our Business Analyst to walk through the requirements and KPIs that guided our work.”



BA – Requirements & KPIs (1 minute)

“Our starting point was clearly defining the business problem: high no-show rates, long wait times, and uneven provider utilization across HealthPulse’s eight clinics.

From this, we created user stories, functional requirements, and acceptance criteria that shaped all analytical and technical decisions. The key KPIs included:

- Reducing the **22% no-show rate** closer to the industry benchmark
- Lowering **45-minute average wait times**
- Improving **provider utilization balance** across clinics
- Providing visibility into operational bottlenecks

These KPIs ensured that the final model, dashboards, and data pipelines all directly supported business outcomes. With the requirements set, the Data Engineering team took over to build the analytical foundation.”

DE – Data Engineering (2 minutes)

“As the Data Engineer on this project, my role was to build a reliable, analysis-ready data environment.

We sourced data from the Snowflake GOLD layer and structured it into a **clean star-schema model**, consisting of:

- **FACT_APPOINTMENTS**
- **DIM_PATIENTS**
- **DIM_PROVIDERS**
- **DIM_CLINICS**
- **DIM_DATES**

I developed a full **Data Dictionary** documenting every field, its datatype, constraints, and business definition. This allowed downstream teams to work consistently and with confidence.

Once the unified dataset was created — around 120,000 appointment records — we handled missing values, standardized clinic and provider identifiers, validated date hierarchies, and ensured that all dimension keys were clean and referentially intact.

The final deliverables included:

- A production-ready star schema
- Cleaned and validated tables
- A data dictionary
- And a consistent dataset handoff for modeling and analytics

This foundation enabled the Data Scientist to build the predictive model and the DA to produce actionable, accurate insights.”

DA – Analytics Insights & Recommendations (2 minutes)

“In analytics, our focus was to uncover operational patterns and provide recommendations that HealthPulse leaders can act on.

We discovered major utilization imbalances across clinics:

- The **Orthopedic Clinic at 173%** and **Urology at 142%** were overstretched
- Meanwhile, **Neurology and Endocrinology** were operating at around **15%**

This mismatch directly contributes to excessive wait times in high-demand clinics while leaving capacity unused elsewhere.

Our recommendation: implement **cross-clinic provider scheduling** and redistribute specialist hours.

We also analyzed day-of-week patterns. Wednesdays and Thursdays consistently carried the heaviest load. We recommended adjusting staffing or incentivizing patients to book on lower-volume days.

Finally, we observed high no-show rates particularly in the **19–40 age group**, followed by 41–65. This led to targeted engagement recommendations:

- App-based and SMS reminders for younger patients
- Calls, emails, and mailed reminders for older groups
- And digital pre-check-in to reduce front-desk congestion

Our Power BI dashboards visualize these findings, showing no-show rates by clinic, provider utilization heatmaps, weekday trends, and financial impacts.”

DS – Predictive Model Overview (2 minutes)

“As the Data Scientist, I developed the no-show prediction model using the dataset prepared by Data Engineering.

We began with exploratory analysis — identifying **lead time**, **appointment time**, **day of week**, **clinic**, and **age group** as strong predictors of no-shows.

For modeling, we built a Logistic Regression pipeline that included:

- One-hot encoding of categorical fields
- Balanced class weights to address the **18% no-show class imbalance**
- A train/test split and evaluation using accuracy, recall, and confusion matrix

The baseline accuracy was around **82%**, but accuracy alone was misleading — the model initially struggled to identify true no-shows due to imbalance. This led us to emphasize recall improvements, class weighting, and the future introduction of more advanced models like Random Forest and XGBoost.

The final deliverables included:

- A full preprocessing + modeling pipeline
- Evaluation reports
- And a saved model file for integration into the API layer

This model will ultimately allow HealthPulse to predict no-shows before they happen and proactively fill or adjust schedules.”

PM – Conclusion (2 minutes)

“To conclude, this one-week sprint delivered a strong foundation for predictive operations at HealthPulse.

We now have:

- A unified analytical dataset
- A documented data model
- Clear insights into clinic performance
- A working predictive model
- And recommendations that directly address revenue loss, provider strain, and patient experience

The next steps are to operationalize the model through the API layer, integrate predictions into scheduling workflows, and pilot test these insights in a live clinic environment.

Thank you — and we’re happy to take questions.”